MILITARY SPECIFICATION

CURTAIN, FLAK PROTECTIVE

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for one type and size of flak protective curtain.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon
UU-P-268	Paper, Kraft, Untreated, Wrapping
DDD-L-20	Label, for Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	Box, Fiberboard
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

Military

MIL-C-508	Cloth, Oxford, Nylon, 3 Ounce
MIL-W-4088	Webbing, Textile, Woven Nylon

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SPECIFICATIONS

Military (Continued)

MIL-T-5038	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-C-7812	Cloth, Nylon, Ballistic
MIL-C-12369	Cloth, Ballistic, Nylon
MIL-G-16491	Grommet, Metallic

STANDARDS

Federal

FED-STD-595	Colors
FED-STD-751	Stitches, Seams, and Stitchings

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality assurance representative. Copies of this specification and other unclassified specifications and standards required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components - The materials and components shall conform to applicable specifications and standards as listed or required herein.

3.1.1 Cloth -

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3.1.1.1 Cover - The cover cloth shall be fabricated from the nylon oxford cloth conforming to MIL-C-508, Type I, Class 1. The color of the cloth shall approximately match FED-STD-595 Olive Drab, Color Number 34087. The dyed cloth shall have at least "good" colorfastness to light and laundering.

3.1.1.2 Filler - The filler cloth shall be fabricated from the ballistic nylon cloth conforming to MIL-C-7812, Type II, or MIL-C-12369 (see 3.2 and Figure 2). The color of the cloth shall be natural.

3.1.2 Frame - The frame shall be fabricated from the nylon webbing conforming to MIL-W-4088, Type XV, 2 inches wide. The color shall approximately match the color of the cover cloth, 3.1.1.1.

3.1.3 Ties - The ties shall be fabricated from the nylon tape conforming to MIL-T-5038, Type III, 3/4 inch wide. The color shall approximately match the color of the cover cloth, 3.1.1.1.

3.1.4 Thread - The thread for all the sewing operations shall be nylon conforming to V-T-295, Type I or II, Class 1 or 2, Size E. The dyed thread shall have at least "good" colorfastness to light and laundering and shall approximately match the color of the cover cloth, 3.1.1.1.

3.1.5 Grommets - The grommets shall be anodized aluminum conforming to MIL-G-16491, Type I, Class 5, Size Number 2.

3.2 Design - The flak protective curtain shall consist of two sets of 5 or 7 plies of the filler cloth (see Figure 2) covered on the visible sides with the cover cloth and attached to a webbing frame containing grommets and tie tapes. The flak protective curtain shall conform in appearance to Figure 1.

3.3 Construction - The details of the construction shall conform to Figures 1 through 3. The cut edges of the nylon cloths, webbings, and tapes shall be seared or treated to prevent fraying. No sharp edges shall be formed. The searing or the method of treatment shall not be deleterious to the cloth, webbing, or tape nor adversely effect any process of the manufacturing. Whenever practical, the selvage edges of the cover and filler cloths shall be utilized in the fabrication of the curtain. The length and width of each cover and filler ply shall not be obtained by joining together two or more pieces of the cloth. The cover and each filler ply shall be one continuous length and width without any seams. The frame shall be fabricated from one continuous length of the webbing. Each tie tape shall be fabricated from one continuous length of the tape.

3.3.1 Seams and stitching - All the seams used in the fabrication of the curtain shall conform to Figures 1 through 3, as applicable. All the regular sewing and the quilting stitching shall be accomplished with Stitch Type 301, conforming to

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FED-STD-751, with 10 to 12 stitches per inch, except for the quilting stitching, which shall contain 4 to 6 stitches per inch. If desired, the edges of each set of filler plies may be overedged stitched together with Stitch Type 502, 503, 504, or 505 conforming to FED-STD-751. The overedge stitching shall contain 4 to 6 stitches per inch and the bight shall be 1/2 inch minimum.

3.3.1.1 Sewing - Each row of stitching shall be straight and parallel to the seam edge or to the previous row of the quilting stitching, as applicable. The straightness of the stitching, in any row, shall be maintained within a tolerance of plus or minus 1/32 inch. The ends of the stitching that are not caught in other seams or stitching shall be backstitched, by overlapping on themselves, a minimum of 1/2 inch. Thread breaks, skips, and runoffs shall be overstitched not less than one inch. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together. No seam shall be twisted, puckered, or pleated and no part of the curtain shall be caught in an unrelated operation or seam. All the thread ends shall be trimmed to a length of 1/4 to 3/8 inch and except for the frame webbing and tie tapes, no raw or cut edges shall be visible. All the seam edges shall be properly forced out and shall not contain any folds.

3.4 Dimensions - The dimensions of the assembled flak protective curtains, width of the seams and hems, size of the quilting pattern, and the dimensions and locations of the components and stitching shall conform to Figures 1 through 3, as applicable, when determined as specified in 4.3.2.

3.5 Weight - When weighed as specified in 4.3.3, the weight of each curtain shall not exceed 10.5 pounds.

3.6 Label - The curtain shall contain an identification label, 1-5/16 by 3 inches, conforming to DDD-L-20, Type I, Class 1, Launderable Label. The label shall be securely sewn on all the four sides in the location specified in Figures 1 and 3.. The label shall contain the following information:

CURTAIN, FLAK PROTECTIVE
MIL-C-18491A(AS)
CONTRACT OR ORDER NO.
NAME OF MANUFACTURER
DATE OF MANUFACTURE (Month and Year)
MAXIMUM WEIGHT 10.5 LBS.

3.7 Workmanship - After completion of the final assembly, the curtains shall be thoroughly cleaned and all loose thread, lint, and foreign matter shall be removed. The flak protective curtains shall not contain any nonspecified hole, tear, cut, mend, burn, needle chew, spot, or stain. Each grommet shall be clinched without distortion, damage, splitting, or cutting of the webbing and shall not contain

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any crack, nick, burr, sliver, sharp edge, corrosion, scale, pit, or dent. Because of the emergency use of this item, the importance of providing a product of uniform excellent quality cannot be overemphasized. The flak protective curtains shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The flak protective curtains shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Visual examination of the flak protective curtains
Dimensional check of the flak protective curtains
Weight of the flak protective curtains
Preparation for delivery

4.2.1 Sampling -

4.2.1.1 Inspection lot -

4.2.1.1.1 Curtain - An inspection lot size shall be expressed in units of one curtain made essentially under the same conditions and from the same materials and components. The sample unit shall be one curtain.

4.2.1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing curtains, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing curtains, fully prepared for delivery with the exception that it need not be sealed.

4.2.1.2 Sampling for the examinations and tests of the curtains and preparation for delivery - The sample size, acceptance criteria, examinations, and tests required for the curtains or the preparation for delivery, as applicable, shall be as specified in Table I.

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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, EXAMINATIONS, AND TESTS OF THE CURTAINS AND PREPARATION FOR DELIVERY

INSPECTION	PARAGRAPH		SAMPLE SIZE	ACCEPTANCE CRITERIA 1/
	REQUIREMENT	METHOD		
Visual examination	3.2 through 3.3.1.1, 3.6, 3.7, and Figures 1 through 3, as applicable	4.3.1.1 and Table II	Inspection Level II	An acceptable quality level of 2.5 defects per 100 units for major defects and 4.0 defects per 100 units for minor defects
Dimensional check	3.4 and Figures 1 through 3, as applicable	4.3.1.1, Table III, and 4.3.2	Inspection Level S-3	An acceptable quality level of 4.0 defects per 100 units
Weight	3.5	4.3.3	Inspection Level S-1	An acceptable quality level of 1.5 defects per 100 units
Preparation for delivery	Section 5	4.3.1.2 and Table IV	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units

1/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

4.3 Inspection methods -

4.3.1 Visual examination -

4.3.1.1 Curtains - Each of the curtains, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and examined visually to determine conformance to this specification. The classification and list of defects, Tables II and III, as applicable, shall be used to classify and enumerate the defects found.

4.3.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing curtains, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table IV, shall be used to enumerate the defects found.

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TABLE II

CLASSIFICATION OF DEFECTS FOR THE VISUAL EXAMINATION OF THE CURTAINS

DEFECT	MAJOR	MINOR
GENERAL		
a. Any non-specified hole, scissors or knife cut, tear, mend, burn, or weakening defect such as smash, multiple float, loose slub, needle chew, or abraded area	X	
b. Color of any component not as specified, not uniform in shade, or contains one or more shade bars		X
c. Any thread end not trimmed to a length of 1/4 to 3/8 inch or loose thread ends		X
d. Any spot or stain		X
e. Any cut edge not seared or treated; any sharp edge formed due to searing or method of treatment		X
f. Any tie tape or frame webbing not one continuous length	X	
g. Number of the filler plies per curtain are less than 10 or 14, as applicable (see 3.2 and Figure 2)	X	
GROMMETS		
a. Any loose, misaligned, distorted, or contains any crack, sharp edge, nick, burr, metal sliver, dent, corrosion, scale, or pit	X	
b. Any surface unclean, contains embedded foreign matter, finish missing, or improperly finished		X
c. Any grommet improperly clinched, resulting in cutting of the webbing	X	
COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of a component or defect of assembly not herein classified	1/	

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TABLE II (Continued)

DEFECT	MAJOR	MINOR
COMPONENTS AND ASSEMBLY (Continued)		
b. Any component, component part, or required operation omitted, or any operation improperly performed not herein classified	1/	
SEAMS AND STITCHING		
a. Accuracy of Seaming:		
1. Any seam, hem, or attachment of any component twisted, puckered, pleated, or caught in an unrelated operation or stitching		X
2. Any joining seam in any cover or filler ply	X	
3. Any seam edge not properly forced out or contains any fold		X
4. Ends of stitching that are not caught in other seams or stitching:		
(a) Not securely backstitched	X	
(b) Secure but backstitched for a distance less than 1/2 inch or the backstitching is not overlapped on itself		X
5. Thread breaks:		
(a) Not securely overstitched	X	
(b) Secure but overstitched for a distance less than one inch		X
b. Gage of stitching 2/:		
1. Any stitching irregular, unevenly gaged, or various stitchings not uniform		X

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TABLE II (Continued)

DEFECT	MAJOR	MINOR
SEAMS AND STITCHING (Continued)		
c. Open seams 3/:		
1. One-fourth inch or less		X
2. More than one-fourth inch	X	
d. Raw edges 4/:		
1. One-half inch or less		X
2. More than one-half inch	X	
e. Seam and stitch type:		
1. Not specified seam or stitch type	X	
f. Stitch tension 5/:		
1. Loose tension, resulting in a loose seam or tight tension, resulting in the breaking of the stitches, when normal pull is applied	X	
g. Stitches per inch 6/:		
1. Stitches more or less than specified:		
(a) One		X
(b) Two or more	X	
LABEL		
a. Missing, incorrect, incomplete, illegible, improperly located, or insecurely attached		X

1/ The defect shall be classified as major, when it seriously affects the appearance or serviceability, otherwise it is to be classified as a minor defect.

2/ The gage of stitching defect shall be scored only, when the condition exists for more than 2 inches on any row of stitching.

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3/ A seam shall be classified as open, when one or more stitches, joining a seam, hem, or quilting, are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be classified as open, when either one or both sides of the seam contain any of the aforementioned for a single stitched seam.

4/ Any edge that should be turned under and is not, but is securely caught in the stitching shall be classified as a raw edge. Any raw edge that is not securely caught in the stitching shall be classified as an open seam.

5/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident, other than the gathering of the material, the stitching shall be inspected by exerting normal pull in the lengthwise direction of the stitching, by pulling taut to straighten out the seam. The gathering of the material shall be classified as an accuracy of seaming defect.

6/ The stitches per inch defect is to be scored only, when the condition exists for more than 3 inches on any row of stitching.

TABLE III

LIST OF DEFECTS FOR THE DIMENSIONS OF THE ASSEMBLED CURTAINS

EXAMINE	DEFECT
Measure the length and width of the assembled curtain, width of the seams and hems, size of the quilting pattern, and the dimensions and locations of the components and stitching.	Any measurement deviating from the dimensions and tolerances as specified in Figures 1 through 3, as applicable, shall be enumerated as a dimensional defect.

TABLE IV

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

ITEM	DEFECT
Exterior and interior markings	Missing, incomplete, incorrect, illegible; of improper size, location, sequence, or method of application; curtain label not visible in polyethylene bag.
Materials	Any non-conforming component; any component, component part, missing, damaged, or otherwise defective.

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TABLE IV (Continued)

ITEM	DEFECT
Workmanship	Inadequate application of the components, such as incomplete closure of the container flaps, loose strapping, etc; bulging or distortion of the containers; polyethylene bag vent hole missing, open or non-continuous heat sealed seams; containers contain metal fastenings or stitches.
Exterior and interior weight or content	Not folded as specified; gross or net weight exceeds the requirement; not individually packaged within the polyethylene bag.

4.3.2 Procedure for measuring - The curtain to be measured shall be laid flat, without tension, on a smooth flat surface so that any creases or wrinkles shall not affect the measurement. The dimensions of the curtain and components, size of the quilting pattern, location of the components and stitching, and width of the seams and hems shall be determined by a yardstick, metal tape, or by any suitable measuring device.

4.3.3 Weight - The weight of the curtain shall be determined on a scale or balance capable of weighing to the nearest 0.1 pound.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A - Each flak protective curtain shall be neatly folded so that the label is on the outerside of the curtain after the final fold. The tie tapes shall be placed on the inside of the folded curtain. The flat folded size of the curtain shall be approximately 12 by 24 inches. Each folded curtain shall be heat sealed within a snug fitting clear polyethylene bag of 0.00125 ±20 percent inch thick material. The heat seal seams shall be straight, continuous, and parallel to each other and the edges of the polyethylene bag. The closing heat seal seam of the bag shall be as close as possible to the edge of the bag. One corner of each bag shall contain a hole, approximately 1/4 inch diameter, to allow the excess air to escape. As an alternate, the final closure of the bag may be accomplished by use of a tuck or reverse flaps.

5.1.2 Level C - The curtains shall be folded and packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

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5.2 Packing - Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A - Six curtains, packaged as specified in 5.1, shall be packed as specified in 5.2.2, except that the fiberboard container shall be Weather-Resistant Class, Variety SW, Grade V3c or V3s. In addition, the container shall be reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

5.2.2 Level B - Six curtains, packaged as specified in 5.1, shall be packed in two equal tiers within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. The top and bottom of the contents of each shipping container shall be completely covered with a sheet of 30 pound minimum basis weight kraft paper conforming to UU-P-268, Grade B. All the seams, corners, and manufacturer's joint shall be sealed with the tape conforming to PPP-T-45, Type III, Grade A, B, or C, 3 inches wide minimum. The container shall be constructed and closed in accordance with the appendix to PPP-B-636. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches.

5.2.3 Level C - The curtains, packaged as specified in 5.1, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The container shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking - In addition to any special marking required by the contract or order, the exterior shipping containers only shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

6. NOTES

6.1 Intended use - The flak protective curtain covered by this specification is intended for protection of aircraft crew members against flak fragments during combat flying.

6.2 Ordering data - Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Quantity desired.

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

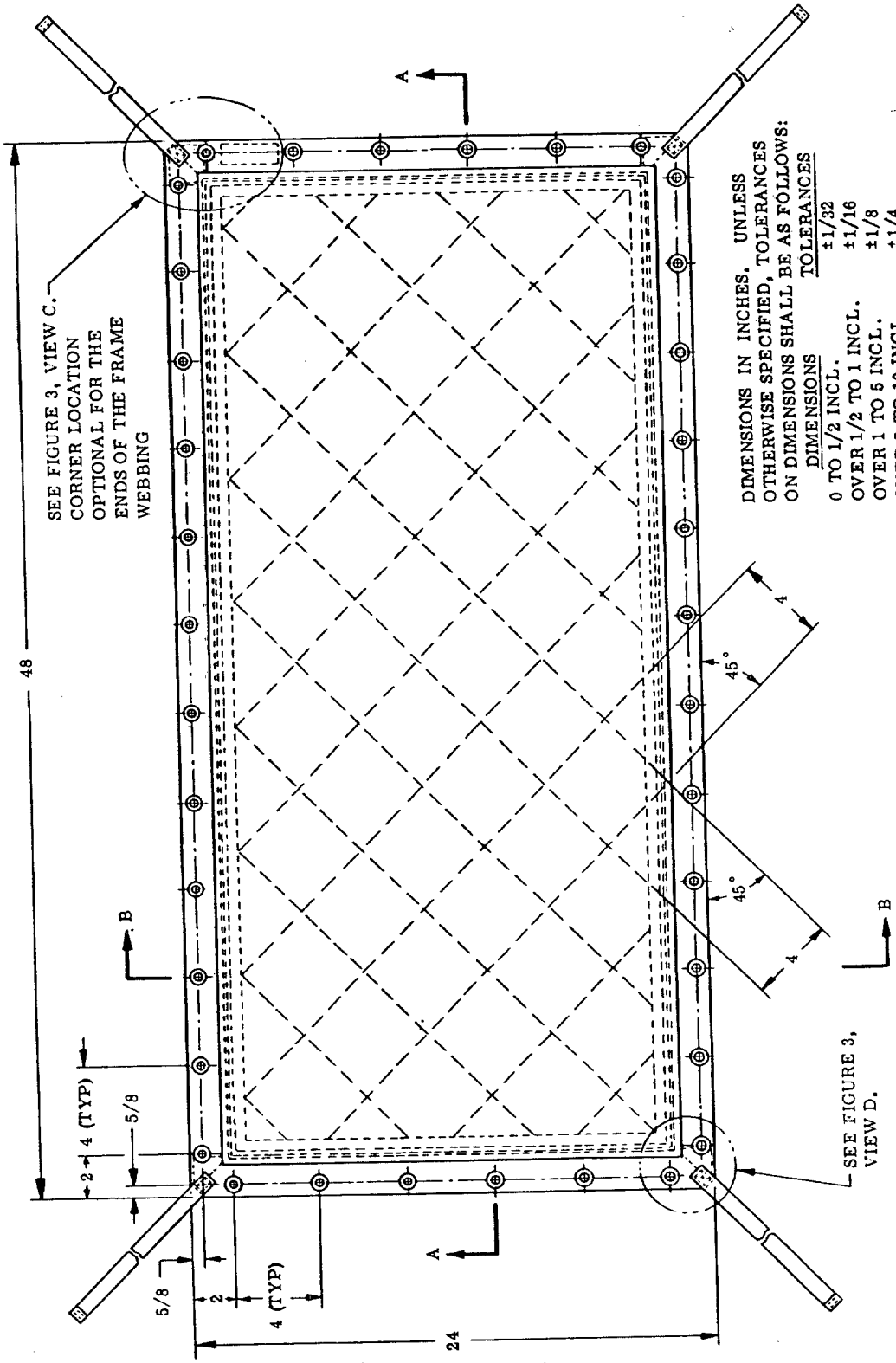
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(d) Whether any special markings are required (see 5.3).

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

Preparing activity:
Navy - AS
(Project No. 1680-N195)

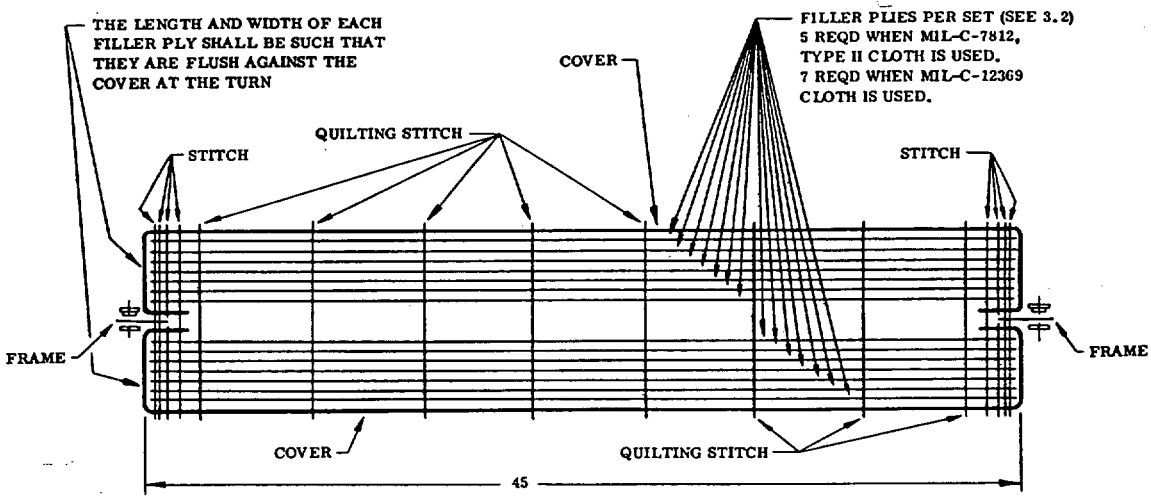
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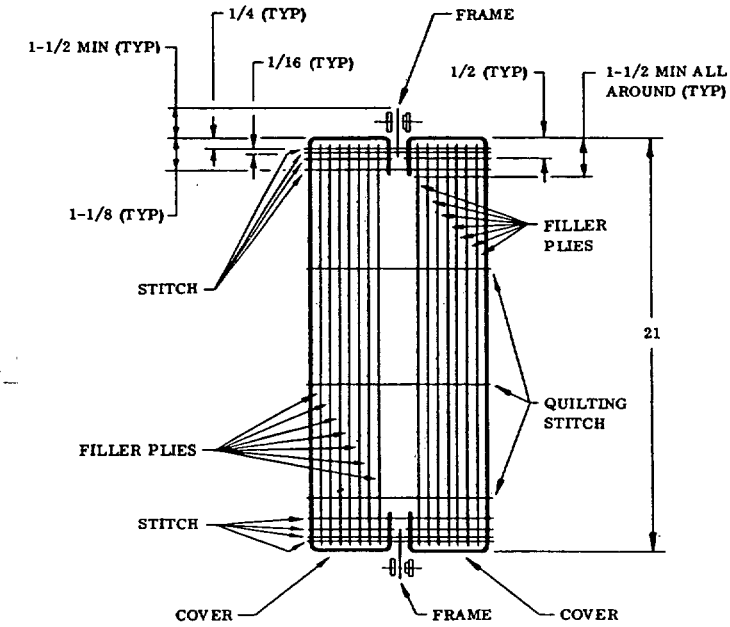


FOR SECTIONS A-A AND B-B, SEE FIGURE 2.

FIGURE 1. FLAK PROTECTIVE CURTAIN

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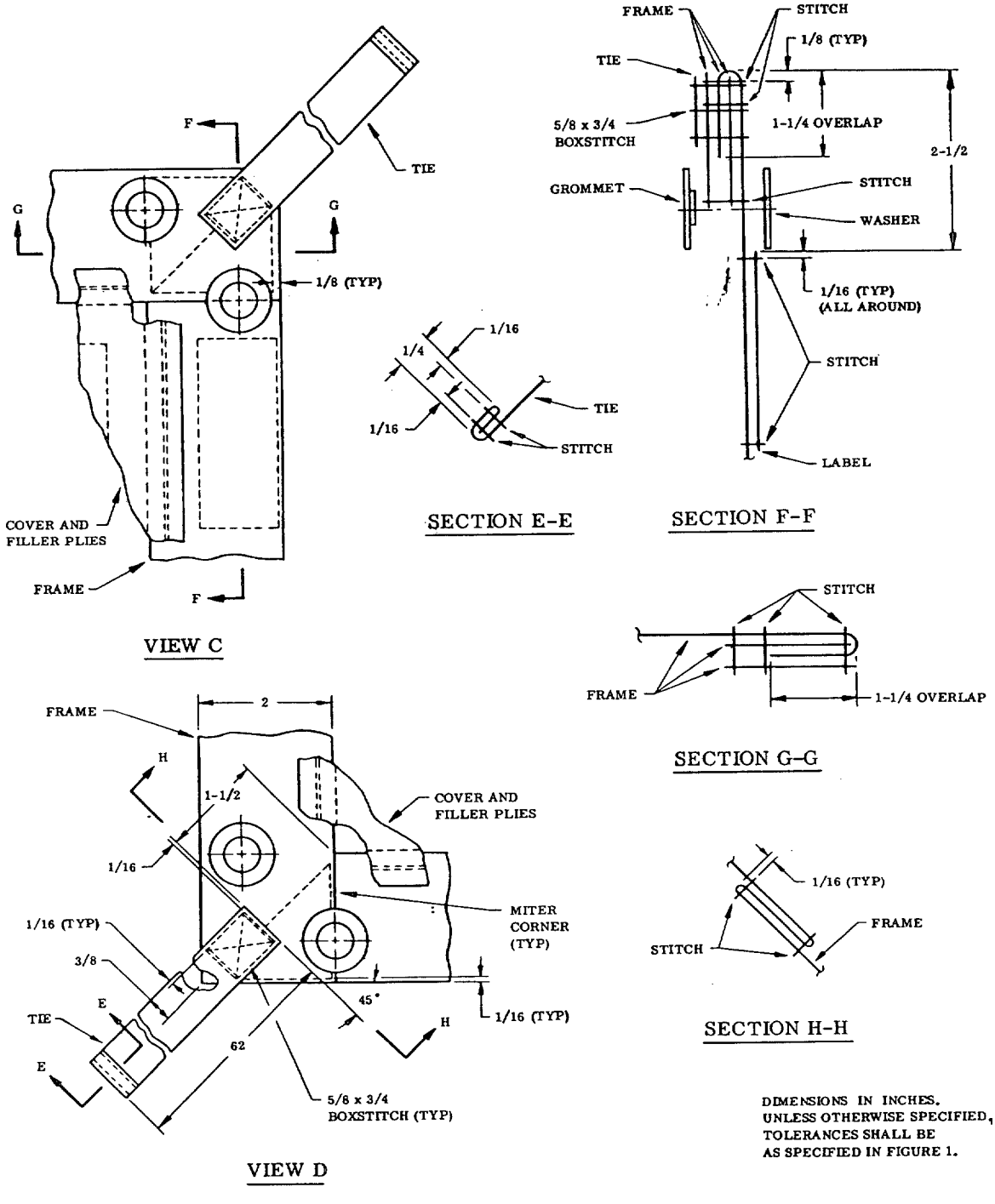




DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

FIGURE 2. FLAK PROTECTIVE CURTAIN SECTIONS

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES SHALL BE
AS SPECIFIED IN FIGURE 1.

FIGURE 3. FLAK PROTECTIVE CURTAIN SECTIONS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

MIL-C-18491A(AS) CURTAIN, FLAK PROTECTIVE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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